Requirements

Roles and their Actions

Guest:

- Sign up, log in and manage your profile
- Search for properties and filter by location, dates and price
- Make, change or cancel bookings
- Complete secure payments
- Message hosts and contact support
- Leave reviews for properties after your stay
- Report incidents

Host:

- Create and manage property listings
- Add amenities and photos to properties
- Set prices, availability and policies
- Handle booking requests
- Message guests
- Review guests after their stay

Administrator:

- Verify users and properties
- Manage platform policies
- Oversee payments and commissions
- Maintain system data

Customer Support:

- Handle support cases
- Resolve disputes between users
- Manage incident reports
- Assist with payment issues

Required Data

Users: UserID, name, email, phone, profile photo, verification status, role-specific attributes.

Properties: Property details, full geographic hierarchy (Country → City → Neighborhood → Address), amenities, photos, availability calendar with pricing.

Bookings: Dates, number of guests, status, total price, cancellation policy reference, links to guest and property.

Payments: Amount, payment method, status, commission, linked to the corresponding booking

Two type of reviews:

- Property reviews: Guests review properties after their stay
- Guest reviews: Hosts review guests after their stay
- Both require completed booking and include rating, comment, date

Messages: Conversations between users, message threads, optional property association.

Support Data: Incident reports, support cases with agents

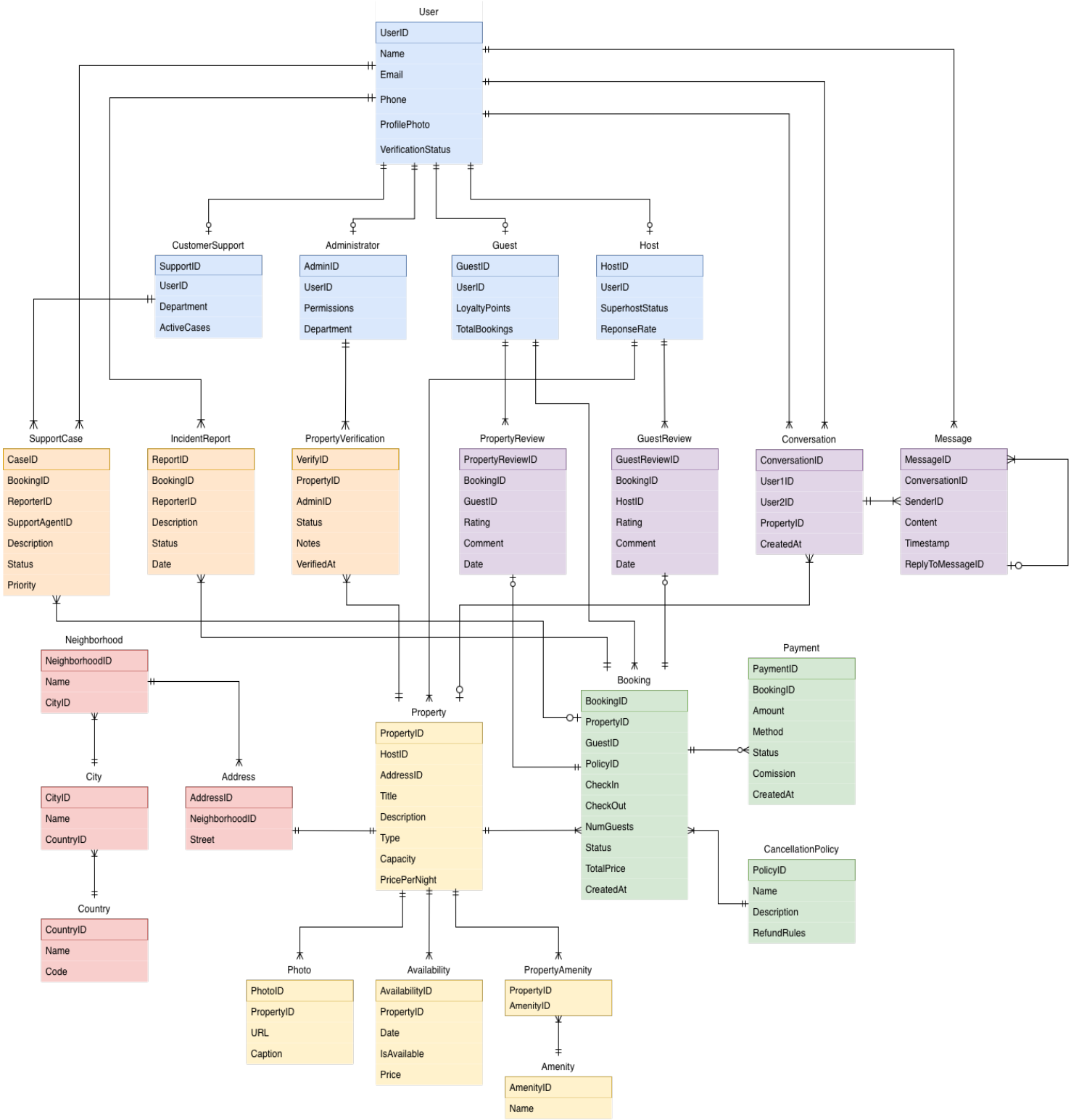
Administrator Data: Cancellation policies, property verification records, list of countries, cities and neighborhoods

Required Functions

Users can create an account, log in and authenticate (guest, host, administrator, support).

- Hosts can list, edit and manage their properties, including potos, amenities and locations details.
- Guests can search for places using filters like dates, prices, location or amenities.
- Calendars show availability and help manage bookings.
- Bookings follow clear rules, including cancellation policies and status updates.
- Payments are processed secure and automatically, including any commissions.
- Hosts and guests can message each other directly within the platform.
- Both guests and hosts can write and read reviews after a stay.
- Users can report incidents and support agents can manage and resolver cases.
- Administrators can verify properties and take care of overall system maintenance to ensure data integrity.

Entity-Relationship Model



Data Dictionary

User Entities

1. User – Base user account

- UserID (INT, PK) – Unique identifier for each user.
- Name (VARCHAR(100)) – User's full name.
- Email (VARCHAR(250)) – Email address used for login and notifications.
- Phone (VARCHAR(20)) – Contact phone number.
- ProfilePhoto (TEXT) – URL of the user's profile picture.
- VerificationStatus (ENUM) – Indicates whether the user is verified.

2. Guest – Users who book properties

- GuestID (INT, PK) – Unique identifier for each guest profile.
- UserID (INT, FK → User) – Links this guest profile to the main user account.
- LoyaltyPoints (INT) – Points earned through bookings.
- TotalBookings (INT) – Number of completed bookings.

3. Host – Users who list properties

- HostID (INT, PK) – Unique identifier for each host profile.
- UserID (INT, FK → User) – Connects the host profile to the user.
- SuperhostStatus (BOOLEAN) – Whether the host holds "superhost" status.
- ResponseRate (DECIMAL(5,2)) – Percentage of messages answered.

4. Administrator – Platform managers

- AdminID (INT, PK) – Unique administrator ID.
- UserID (INT, FK → User) – Links administrator rights to a user.
- Permissions (TEXT) – List of administrator permissions.
- Department (VARCHAR(100)) – Administrator department.

5. CustomerSupport – Support staff

- SupportID (INT, PK) – Unique ID for each support agent.
- UserID (INT, FK → User) – Connects customer support role to the user.
- Department (VARCHAR(100)) – Support team.
- ActiveCases (INT) – Number of open cases.

Properties Entities

6. Property – Rental listings

- PropertyID (INT, PK) – Unique identifier for each listing.
- HostID (INT, FK → Host) – Owner or manager of the property.
- AddressID (INT, FK → Address) – Linked location details.
- Title (VARCHAR(200)) – Listing title shown to guests.
- Description (TEXT) – Detailed description of the property.
- Type (VARCHAR(50)) – Property type.
- Capacity (INT) – Maximum number of guests allowed.
- PricePerNight (DECIMAL(10,2)) – Standard nightly rate.

7. Amenity – Property features

- AmenityID (INT, PK) – Unique amenity identifier.
- Name (VARCHAR(100)) – Amenity name.

8. PropertyAmenity – Links properties to amenities

- PropertyID (INT, FK → Property) – Property linked to the amenity.
- AmenityID (INT, FK → Amenity) – Amenity linked to the property.

  Composite PK: (PropertyID, AmenityID)

9. Photo – Property images

- PhotoID (INT, PK) – Unique identifier for the photo.
- PropertyID (INT, FK → Property) – Property the photo belongs to.
- URL (TEXT) – Location of the image.
- Caption (VARCHAR(250)) – Short image description.

10. Availability – Booking calendar

- AvailabilityID (INT, PK) – Unique availability ID.
- PropertyID (INT, FK → Property) – Property reference.
- Date (DATE) – Calendar date.
- IsAvailable (BOOLEAN) – Whether the property is free that day.
- Price (DECIMAL(10,2)) – Price for this specific date.

Booking Entities

11. Booking – Reservations

- BookingID (INT, PK) – Unique booking identifier.
- PropertyID (INT, FK → Property) – Reserved property.
- GuestID (INT, FK → Guest) – Guest making the reservation.
- PolicyID (INT, FK → CancellationPolicy) – Cancellation rules applied.
- CheckIn (DATE) – Arrival date.
- CheckOut (DATE) – Departure date.
- NumGuests (INT) – Number of guests included.
- Status (ENUM) – Booking status (pending, confirmed, cancelled).
- TotalPrice (DECIMAL(10,2)) – Total amount before payment.
- CreatedAt (DATETIME) – Timestamp when booking was made.

12. Payment – Financial transactions

- PaymentID (INT, PK) – Unique payment ID.
- BookingID (INT, FK → Booking) – Booking that the payment belongs to.
- Amount (DECIMAL(10,2)) – Charged amount.
- Method (VARCHAR(50)) – Payment method.
- Status (ENUM) – Payment status (pending, paid, failed).
- Commission (DECIMAL(10,2)) – Platform commission.
- CreatedAt (DATETIME) – Time of transaction.

13. CancellationPolicy – Refund policies

- PolicyID (INT, PK) – Unique cancellation policy ID.
- Name (VARCHAR(100)) – Policy name.
- Description (TEXT) – Policy details.
- RefundRules (TEXT) – Refund conditions.

Communication Entities

14. PropertyReview – Reviews of properties

- PropertyReviewID (INT, PK) – Unique review identifier.
- BookingID (INT, FK → Booking) – Booking linked to the review.
- GuestID (INT, FK → Guest) – Reviewer.
- Rating (DECIMAL(3,2)) – Rating score.
- Comment (TEXT) – Review text.

- Date (DATE) – Review date.

15. GuestReview – Reviews of guests

- GuestReviewID (INT, PK) – Unique review identifier.
- BookingID (INT, FK → Booking) – Linked booking.
- HostID (INT, FK → Host) – Reviewer author.
- Rating (DECIMAL(3,2)) – Rating score.
- Comment (TEXT) – Review text.
- Date (DATE) – Review date.

16. Conversation – User chats

- ConversationID (INT, PK) – Chat thread identifier.
- User1ID (INT, FK → User) – First participant.
- User2ID (INT, FK → User) – Second participant.
- PropertyID (INT, FK → Property) – When the chat is about a property. It can be null.
- CreatedAt (DATETIME) – When the conversation started.

17. Message – Individual messages

- MessageID (INT, PK) – Unique message identifier.
- ConversationID (INT, FK → Conversation) – Chat the message belongs to.
- SenderID (INT, FK → User) – Sender of the message.
- Content (TEXT) – Message text.
- Timestamp (DATETIME) – Send time.
- ReplyToMessageID (INT, FK → Message) – Message being replied to.

<u>Geography Entities</u>

18. Country – Countries

- CountryID (INT, PK) – Unique country identifier.
- Name (VARCHAR(100)) – Country name.
- Code (VARCHAR(15)) – Country code.

19. City – Cities

- CityID (INT, PK) – Unique city identifier.
- Name (VARCHAR(150)) – City name.
- CountryID (INT, FK → Country) – Country the city belongs to.

20. Neighborhood – Neighborhoods

- NeighborhoodID (INT, PK) – Unique neighborhood identifier.
- Name (VARCHAR(150)) – Neighborhood name.
- CityID (INT, FK → City) – City the neighborhood belongs to.

21. Address – Property locations

- AddressID (INT, PK) – Unique address identifier.
- NeighborhoodID (INT, FK → Neighborhood) – Neighborhood where the property is located.
- Street (VARCHAR(200)) – Street name and number.

<u>Management Entities</u>

22. IncidentReport – Issue reports

- ReportID (INT, PK) – Unique incident report identifier.
- BookingID (INT, FK → Booking) – Booking involved.
- ReporterID (INT, FK → User) – User submitting the report.
- Description (TEXT) – Explanation of the issue.
- Status (ENUM) – Status of the report.
- Date (DATETIME) – Report date.

23. SupportCase – Support

- CaseID (INT, PK) – Support case identifier.
- BookingID (INT, FK → Booking) – Linked booking. It can be null.
- ReporterID (INT, FK → User) – User who created the case.
- SupportAgentID (INT, FK → CustomerSupport) – Customer support agent handling the case.
- Description (TEXT) – Case details.
- Status (ENUM) – Case status.
- Priority (ENUM) – Priority level.

24. PropertyVerification – Property approvals

- VerifyID (INT, PK) – Verification record identifier.
- PropertyID (INT, FK → Property) – Property being verified.
- AdminID (INT, FK → Administrator) – Administrator who verified it.
- Status (ENUM) – Verification status.
- Notes (TEXT) – Additional comments.
- VerifiedAt (DATETIME) – Time of verification.

Data Types:

- INT: Whole numbers
- VARCHAR: Text (length varies)
- TEXT: Long text
- DECIMAL: Numbers with decimals
- DATE: Date only
- DATETIME: Date and time
- ENUM: Predefined values
- BOOLEAN: True/False
- PK: Primary Key (unique identifier)
- FK: Foreign Key (links to another table)